THE PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT MID TERM JOINT EXAMINATION

FORM THREE
031 PHYSICS 1

Time: 3 Hours **Monday, 31st August 2026 a.m.**

Instructions

1. This paper consists of sections A, B and C with a total of **eleven (11)** questions
2. Answer **all** questions in sections A and B and **two (2)** questions from section C in your answer sheet provided
3. Section A carries **sixteen (16)** marks, section B **fifty four (54)** and C carries **thirty (30)** marks
4. Cellular phones and any unauthorized materials are not allowed in the examination room
5. Non-programmable calculators and mathematical tables may be used
6. Where necessary the following constants may be used:
   i. Acceleration due to gravity, $g = 10m/s^2$
   ii. Density of water $= 1.0g/cm^3$
   iii. Pie, $\pi = 3.14$

SECTION A (16 MARKS)

Answer **all** questions in this section.

1. For each of the items (i) – (x), choose the correct answer from among the given alternatives and write its letter beside the item number in the answer sheet provided.

   i. Which one of the following arguments describes the mathematical language used in Physics?
   A. Matter occupies space
   B. Density is mass per volume
   C. Volume is the amount of space occupied by the body
   D. Physics is a branch of science.
   E. Physics is broad

   ii. A physics teacher asked the students to explain briefly things that may cause fire in the physics laboratory. The students tried to answer but some explanations were not correct. Identify the explanations which were not correct
   A. Electric faults.
   B. Presence of flammable materials.
   C. Carelessness in using gas lighter and match boxes.
   D. The color of some chemicals supports combustion.
   E. Presence of corrosive materials

iii. A car moving at steady speed has a frictional force on its surface whose size depends on?
   A. Its speed and surface area
   B. Its speed only.
   C. Its area only.
   D. Its weight.
   E. Its wheel speed only.

iv. In loading a lorry, a man lifts boxes each of weight 100N through a height of 1.5m. if he lifts 4 boxes per minute, the average power the man is working is:
   A. 100W
   B. 10W
   C. 600W
   D. 600J
   E. 2250W

v. Which statement about the properties of gaseous state of matter is true?
   A. Particles vibrate in fixed positions
   B. The shape is indefinite with definite volume
   C. Particles are closely packed together
   D. Both shape and volume are definite
   E. The inter-particle distance is large.

vi. An object has a mass of 5kg. What is its kinetic energy when it is moving at speed of 10m/s?
   A. 50J
   B. 250J
   C. 150J
   D. 100J
   E. 35J

vii. A bus carrying a very big load on its carrier can easily overturn because
   A. It cannot run fast
   B. Its equilibrium is stable
   C. Its center of gravity is low
   D. It is more stable and can run faster
   E. Its center of gravity is high

viii. In refraction of light through a prism, the light ray
   A. Suffers refraction only at one face of the prism.
   B. Emerges out from the prism in a direction parallel to the incident ray.
   C. Bends at both the surface of the prism towards its base.
   D. Bends at both surfaces of prism opposite to its base.
   E. Reflects back to its own path.

ix. A car moving round a circular path at constant speed. Which of its motion properties is constant throughout?

A. Displacement
B. Velocity
C. Acceleration
D. Momentum
E. Kinetic energy

x. The greatest pressure exerted on a smooth horizontal surface by the surface of a regular block of wood of dimensions $3cm \times 4cm \times 5cm$ and the density of 0.75g/cm^3 is:
A. 225N/m^2
B. 300 N/m^2
C. 150 N/m^2
D. 375 N/m^2
E. 275 N/m^2

2. Match each of the descriptions of the terms used in simple machines in LIST A with the corresponding concept used in simple machines in LIST B**.**

List A	**List B**
(i) The ratio of the distance moved by effort to the distance moved by the load.	A. A simple pulley
(ii) The ratio of the load raised steadily by a machine when an effort or force is applied	B. Combination pulley
(iii) A fixed wheel with a rope passing round a groove in the wheel's circumference	C. Efficiency
(iv) The ratio of the work output to the work input times 100%	D. Lever
(v) Consists of a rigid bar that moves about a fixed point.	E. Mechanical Advantage
(vi) First class lever	F. Single fixed pulley
	G. The block and tackle pulley system
	H. Velocity ratio.
	I. F-L-E

SECTION B (54 MARKS)

Answer **all** question

3. (a) (i) If a pulley system is 100% efficient, its mechanical advantage is numerically equal to its velocity ratio. However, the efficiency of a pulley system is always less that 100%. Explain with two reasons why so.
(ii) Draw an ideal pulley system with a velocity ratio of 4.
(b) A certain first class lever of length 2.5 m has a velocity ratio of 12 and efficiency of 85%
(i) How far from the fulcrum, is the effort applied?
(ii) What effort is required to lift a load weighing 75N?

4. (a) Describe the evidence that shows that molecules in gases are in a state of motion.
(b) Two forces of 6 N and 8 N act at right angles to each other on an object.
(i) Draw a vector diagram representing the two forces. **(1 mark)**
(ii) Calculate the magnitude of the resultant force. **(2 marks)**

5. One form of a hydrometer is used to measure the densities of liquid over the range of 0.8 to 1.00 g/cm^3. If the area of cross section of its stem is 0.5 cm^2, and the distance between the 0.80 and 100 division is 18 cm. Determine:
   (a) The volume of the hydrometer below 1.00 graduation
   (b) The position of the 0.90 graduation

6. (a) Explain why it would not be practical to make a Fortin type barometer with water instead of mercury.
   (b) A student uses a simple magnifying glass to observe a small object.
      (i) State the type of lens used in a magnifying glass.
      (ii) Where should the object be placed relative to the focal point of the lens to obtain magnification?
      (iii) State two characteristics of the image formed by a magnifying glass.

7. (a) Why does the sky appear blue while being observed from the Earth and black when you are on the Moon?
   (b) When a screw gauge with a least count of 0.01mm is used to measure the diameter of a rod, the reading on the sleeve is found to be 1.6cm and the reading on the thimble is found to be 48 divisions. What is the correct diameter of the rod, if the zero error for the gauge is -0.003cm?

8. (a) How is the electroscope used for testing the types of charges?
   (b) After walking across a carpeted floor, you sometimes get a mild electric shock when you touch a metal doorknob. Explain how this happens.

SECTION C (30 MARKS)

Answer any **two** question

9. (a) Explain how heat is transferred from one region to another by:
      (i) Conduction.
      (ii) Convection.
      (iii) Radiation.
   (b) It is difficult to unscrew wheel nuts in the morning, while it is relatively easy to unscrew them on a hot day.
   (c) What happens to the same ray of light within the diamond when the diamond is surrounded by water?

10. (a) Friction force is necessary evil; explain in five points.
   (b) When two surfaces are polished, the coefficient of friction between them decreases. However, the coefficient of friction increases and becomes very large if the surfaces are made highly smooth. Explain
   (c) A block of wood of 4kg just slides without acceleration down an inclined plane of 40^0 to the horizontal. What is the coefficient of dynamic friction?

11. (a) State the rules used to locate image on the curved surface

(b) To a diver under water, most of the surface looks silvery. Bubbles of air rising from the diver look silvery. Explain

(c) An object is placed 20 cm in front of a concave mirror of focal length 12 cm. Find the position and the nature of the image formed.